

ECx00M&ECx00N&EC800K Series

PPP Application Note

LTE Standard Module Series

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About the Document

Revision History

Version	Date	Author	Description
-	2020-03-09	Chaim QIAN	Creation of the document
1.0	2023-05-16	Chaim QIAN	First official release
1.1	2023-09-25	Chaim QIAN	1. Added the applicable module EC800K series. 2. Added a note that different models support PPP feature differently due to the flash size of the module (Chapter 1.1). 3. Deleted the note about CDMA network (Chapter 3.1).

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1 Introduction

This document explains the PPP function of Quectel ECx00M family, ECx00N family and EC800K series modules, including PPP application modes, setting procedures, connection modes, termination steps, and dialing examples.

1.1. Applicable Modules

Table 1: Applicable Modules

Module Family	Module
ECx00M	EC200M-CN
	EC600M-CN
	EC800M-CN
ECx00N	EC200N-CN
	EC800N-CN
-	EC800K Series

NOTE

Limited by the flash size of the module, different models of the modules support PPP feature differently. For more details, please contact Quectel Technical Support.

2 PPP Application Mode

The use of PPP (Point-to-Point Protocol) is illustrated in the following figure. Use UART or USB port for PPP connection. The module provides a PPP server for application, and the application provides a PPP client for the module. Meanwhile, the application must provide protocols such as TCP/IP, HTTP(S), etc. After the PPP connection is established, the IP packet data stream from the application will be transmitted to the Internet through the module.

Most standard operating systems (e.g., Windows, Unix/Linux) include the PPP protocol stack. For operating systems that do not have applications for PPP connection, it is necessary to develop the relevant application software first.

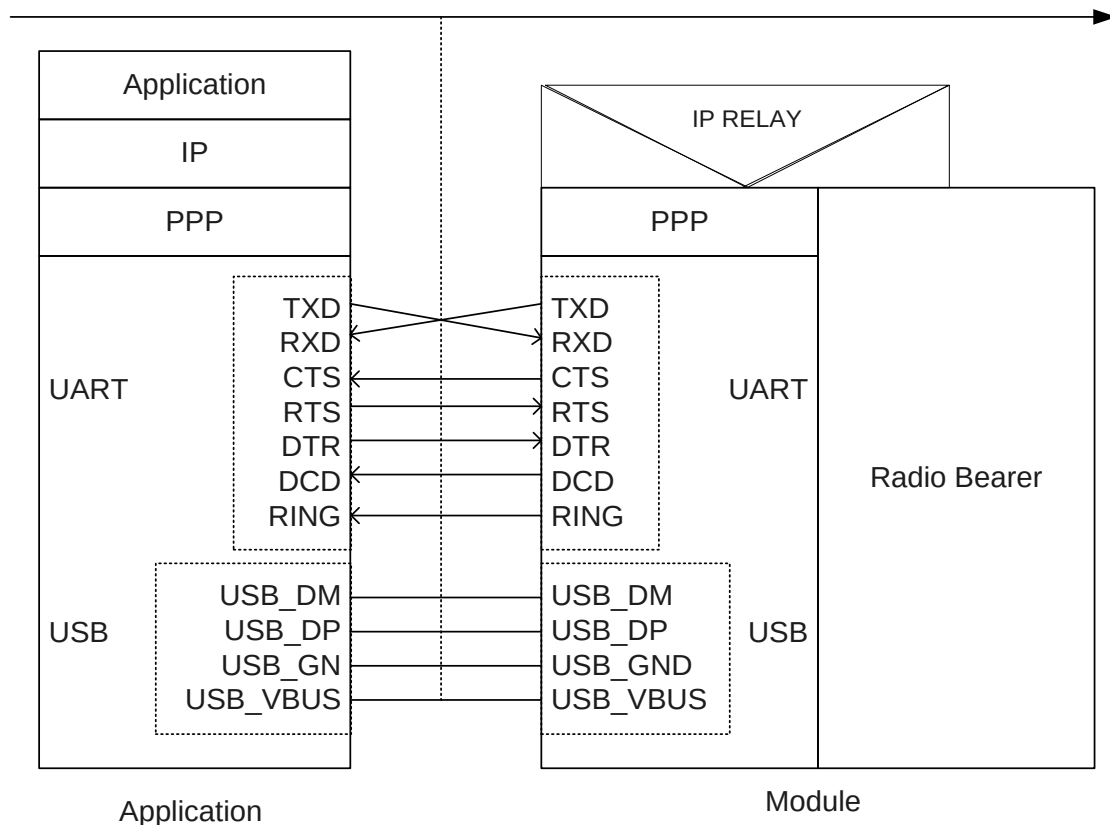


Figure 1: PPP Application Mode

3 PPP Link Establishment

3.1. General Procedures for PPP Link Establishment

After the module registers on GPRS network, it is necessary to set the APN for PPP with **AT+CGDCONT** and to start PPP with **ATD*99#**. After executing **ATD*99#**, the module will enter the PPP frame interaction process, which is based on standard PPP protocol. The module data packet interaction is described in the figure below. For more information about standard PPP, see *RFC 1661*.

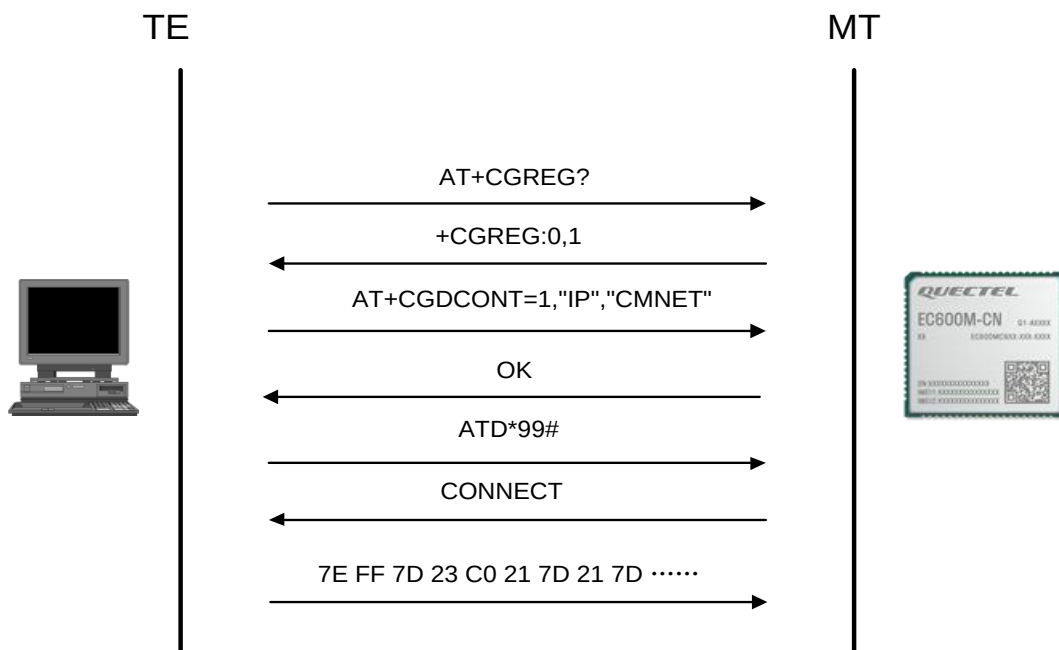


Figure 2: General Procedures for PPP Link Establishment

3.2. Recommended Procedures for PPP Link Establishment

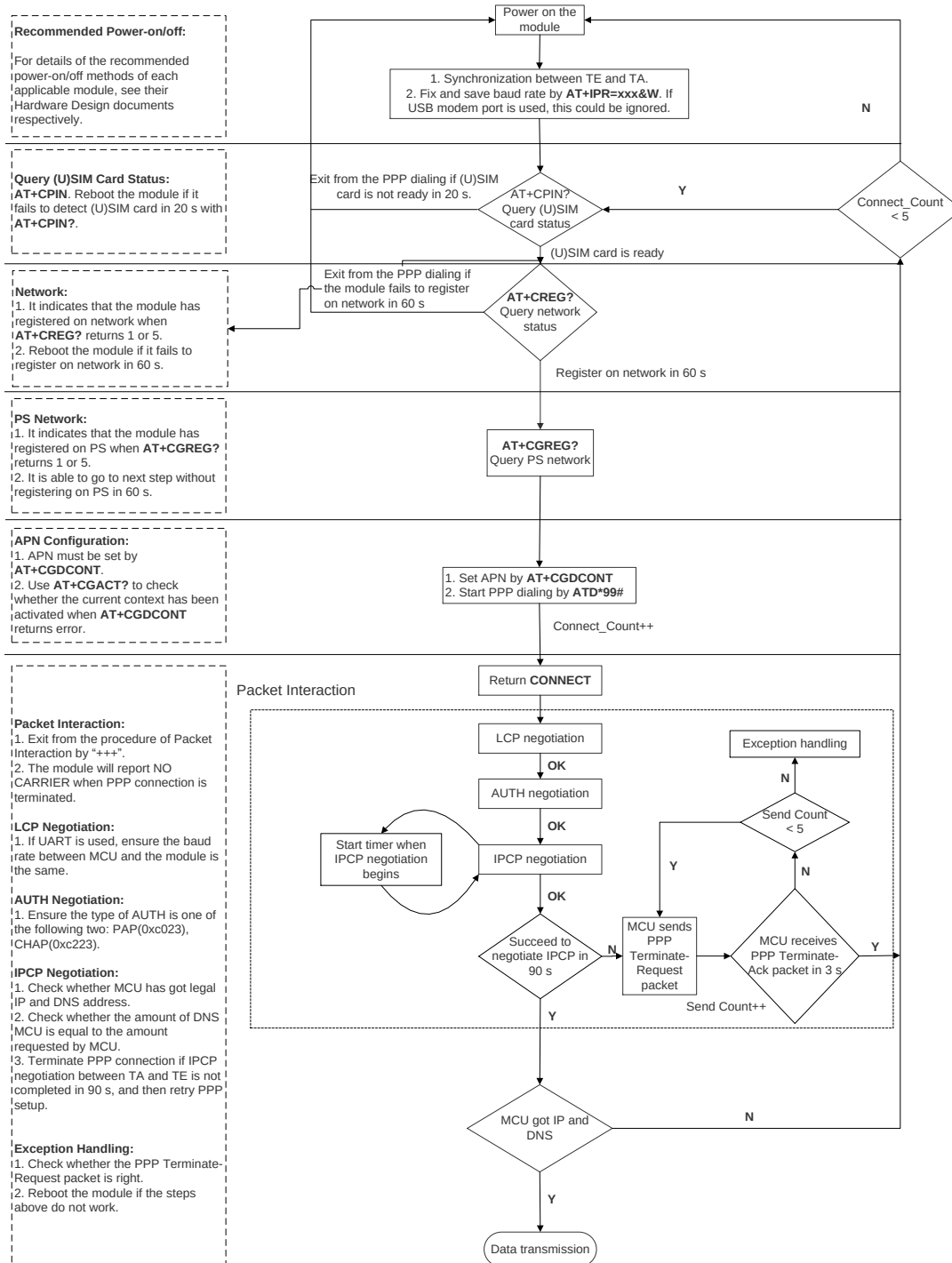


Figure 3: Recommended Procedures for PPP Link Establishment

When the module is powered on, if the main UART is used, set the baud rate by executing **AT+IPR=xxx;&W**. Before executing **ATD*99#** to start PPP dialing, you must check the status of the (U)SIM card with **AT+CPIN?**. When the (U)SIM card is ready, periodically check the network registration status with **AT+CREG?** and **AT+CGREG?** until the network is ready.

NOTE

1. Ensure that the MCU and the module are synchronized after rebooting the module. MCU sends **AT<CR><LF>** to the module every 100 ms until it receives **OK** reported by the module. If the main UART is used, the MCU will execute **AT+IPR=xxx;&W** to set and save the baud rate after synchronizing the baud rate.
2. Before inputting the next AT command, the MCU needs to wait for the response to the AT command (such as response **OK**, **CME error**, **CMS error**) that is being executed. If there is no response within 60 seconds, the module can be restarted.
3. Do NOT power on/off the module frequently. If 3 consecutive dialing attempts fail, you can immediately power on/off (reset) the module for the first time. After resetting, if the dialing attempt fails, you need to wait 10 minutes for the second reset, 30 minutes for the third reset, and one hour for the fourth reset.
4. If the MCU fails to transmit data to network after establishing the PPP connection, check the PPP configuration and network status and restart the module.

4 PPP Connection Modes

4.1. Data Mode and Command Mode

The module communicates with an application via a USB or UART port (including AT commands and data). USB and UART ports have two working modes: data mode and command mode.

The two ports are in command mode before PPP is set, and the module can execute AT commands in this mode. When PPP negotiation starts, the two ports enter data mode, and remain in the mode once the PPP connection is established. If the PPP connection is not successfully established, the two ports will be in the command mode. In data mode, the module cannot execute AT commands.

ECx00M family, ECx00N family and EC800K series modules provide convenient methods to switch between the two modes.

4.1.1. Switch from Data Mode to Command Mode

4.1.1.1. Change DTR Level to Switch from Data Mode to Command Mode

When the PPP connection has been established and USB/UART port is in data mode, change the DTR pin level from low to high to switch from data mode to command mode (**AT&D1** must be set first). After a successful switch, the module will return **OK**.

4.1.1.2. Use Sequence +++ to Switch from Data Mode to Command Mode

After the PPP connection is established, use **+++** to switch the USB/UART port from data mode to command mode. To prevent the **+++** escape sequence from being misinterpreted as data, the following requirements should be followed:

- 1) Do not input any character for at least 1 s before inputting **+++**.
- 2) Input only **+++** within 1 s, USB/UART port will switch from data mode to command mode, and the module will return **OK**.
- 3) Do not input any character for within 1 s after inputting **+++**.

NOTE

Perform the above operations after completing the PPP negotiation, otherwise the above operations will terminate the PPP negotiation and the USB/UART port will exit the data mode. After completing the PPP negotiation, when the USB/UART port switches to the command mode, the input data will be treated as AT command, and the module maintains the PPP connection.

4.1.2. Switch from Command Mode to Data Mode

4.1.2.1. Use ATO to Switch from Command Mode to Data Mode

Execute **ATO** to switch USB/UART port from command mode to data mode, once the PPP connection has been established.

Example

//Once the PPP connection has been established, USB/UART port is in command mode.

ATO

CONNECT //Indicates that TA has entered data mode, and all data inputted via USB/UART port will be treated as PPP frames.

4.2. Handle URC in Data Mode

The URC for incoming calls and short messages in data mode will not be reported to the PPP dial-up port during an ongoing PPP connection. However, the level of module's RI pin will change from high to low for 120 ms as an indication of URC for incoming calls and short messages. The MCU can switch the port to command mode according to the RI pin status to process incoming calls or short messages. After switching to command mode, the URC will be reported to the PPP dial-up port if there are still incoming calls or text messages.

4.3. Data Carrier Detection (DCD) Mode

DCD mode is determined by **AT&C**. If **AT&C0** is executed, the DCD pin will not be used to indicate the data carrier status. If **AT&C1** is executed, the DCD pin will be used to indicate the data carrier status. The pin will remain at low level if there is a data carrier or PPP negotiation begins, otherwise it will remain at high level.

NOTE

When switching module's USB/UART port from data mode to command mode (by using **+++**), the DCD state remains unchanged.

5 PPP Connection Termination

There are two methods to terminate the PPP dial-up connection:

1. Terminating the PPP connection through the LCP Terminate-Request message – recommended method.
2. Terminating the PPP connection by changing the DTR level. Set the DTR function with **AT&D2**, change the DTR level from low to high, and the data connection will be automatically terminated. Once the PPP connection is terminated, the USB/UART port will enter the command mode.

Example

//USB/UART port is in command mode before PPP connection is established.

AT&D2

OK

NOTE

1. PPP connection termination procedures must be performed in data mode.
2. PPP connection termination procedures can be performed at any time during the PPP setting or connecting process.

6 PPP Dial-up Operations

This chapter explains how to establish a PPP dial-up connection in Windows 10. For detailed operations in Linux operating system, see **document [1]**.

6.1. Preparation

It is necessary to perform the following steps before establishing a PPP dial-up connection in Windows.

1. Connect the module to a PC and enter the PIN code if the (U)SIM card PIN is locked.
2. Make sure the (U)SIM card can successfully register on GPRS network.

6.2. PPP Dial-up in Windows 10

6.2.1. Modem Configuration

6.2.1.1. Add New Modem

If the **Standard 19200 bps Modem** is not installed, add a new standard modem to the modem section of the control panel.

1. Search for “**Phone**” in the main interface, and then click “**Phone and Modem**” option:

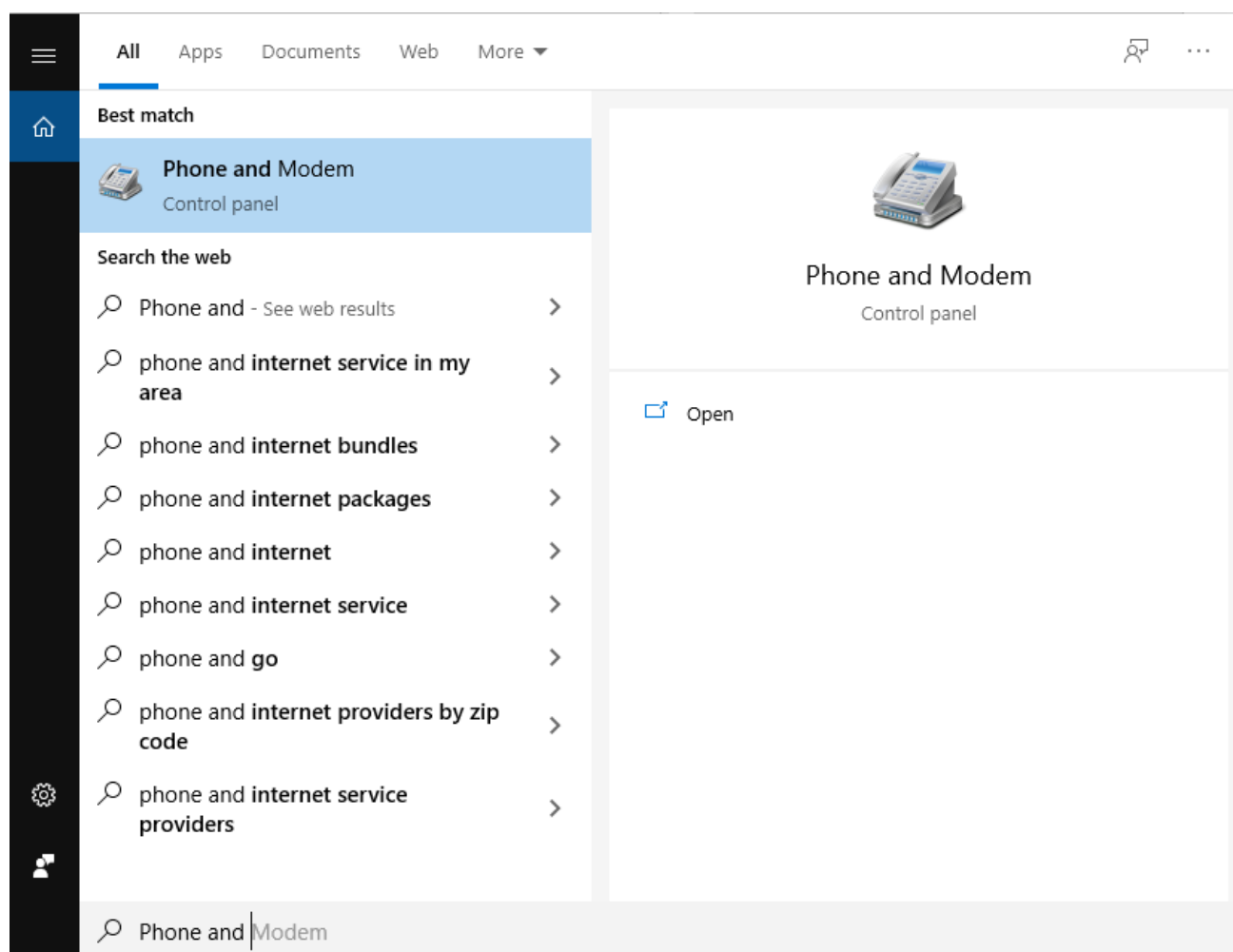
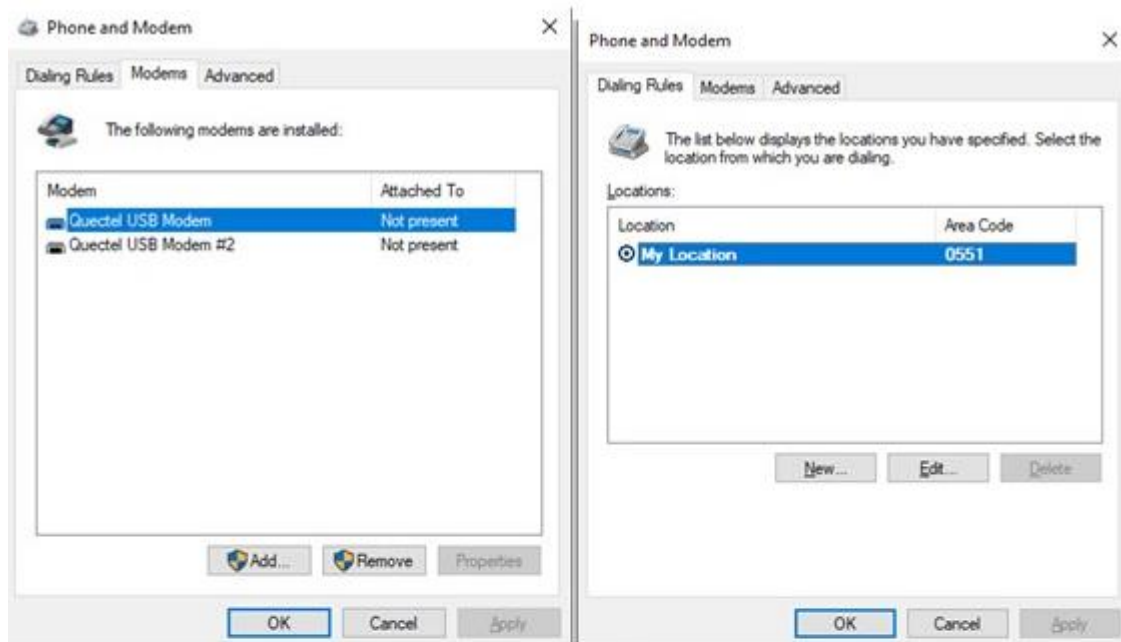


Figure 4: Search for “Phone and Modem” in Control Panel

2. Double-click the “**Phone and Modem**”, and select “**Modems**” → “**Add...**” to add a new modem.



Add Hardware Wizard

Install New Modem

Do you want Windows to detect your modem?

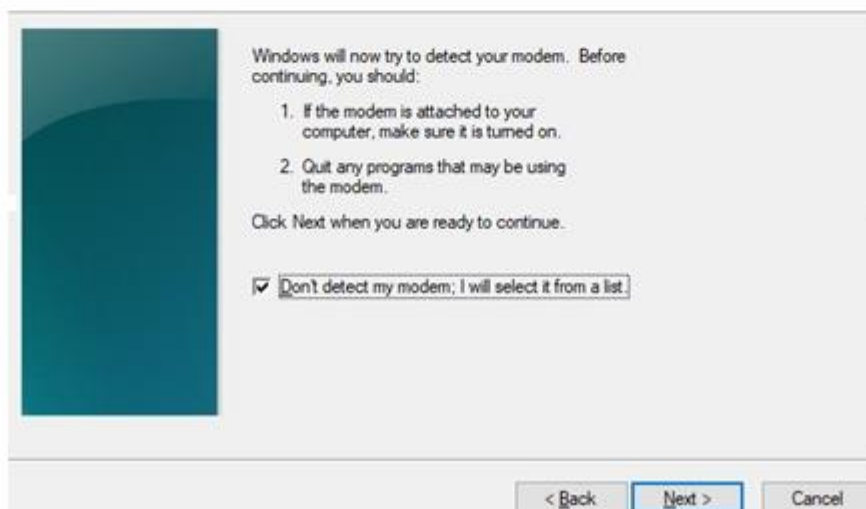


Figure 5: Add New Modem

- Follow the instructions on the screen to install the new modem: select “**Standard 19200 bps Modem**” and then the port (such as “**COM10**”) to be installed; click “**Next**” button until the configuration is complete. For details, see the figures below.

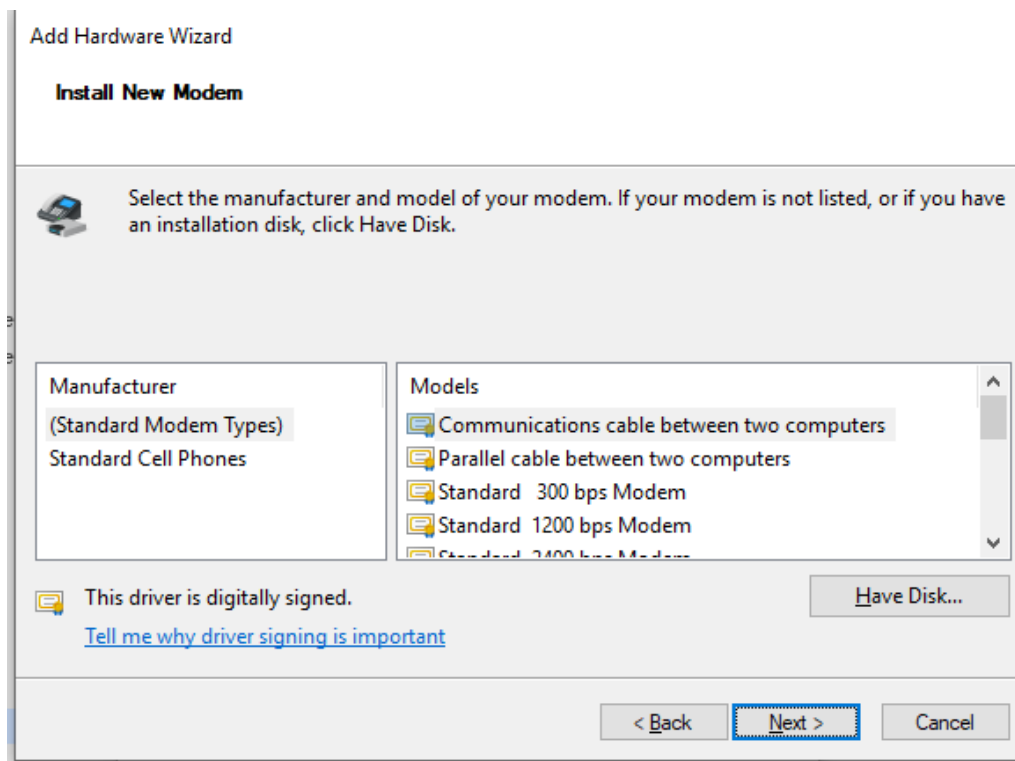


Figure 6: Select Modem Model

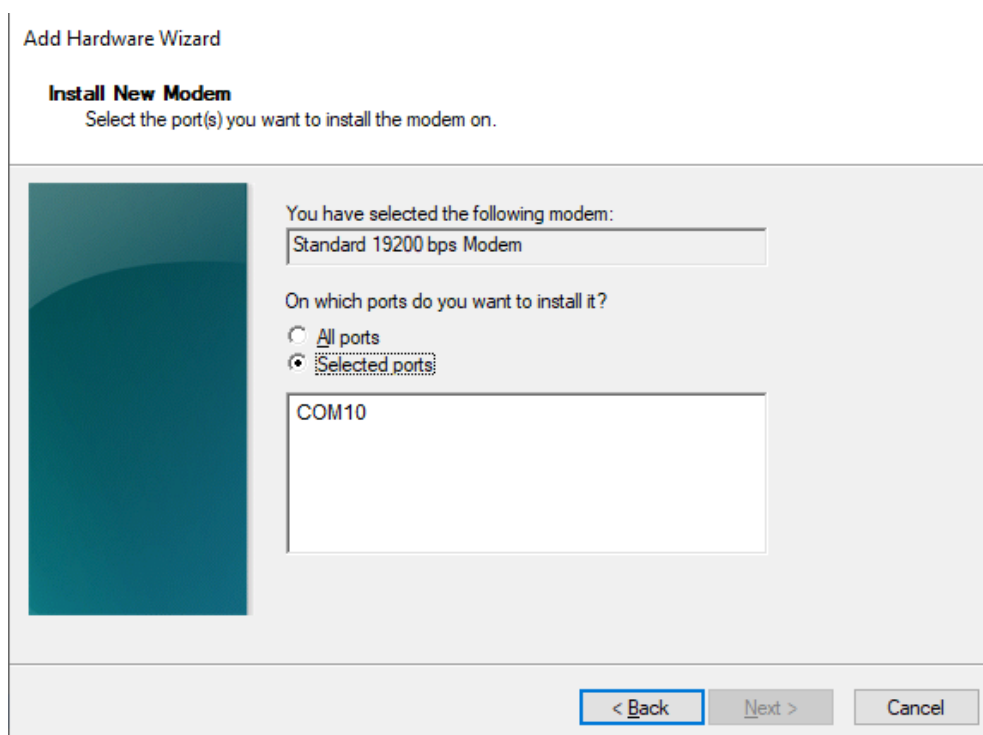


Figure 7: Select Port

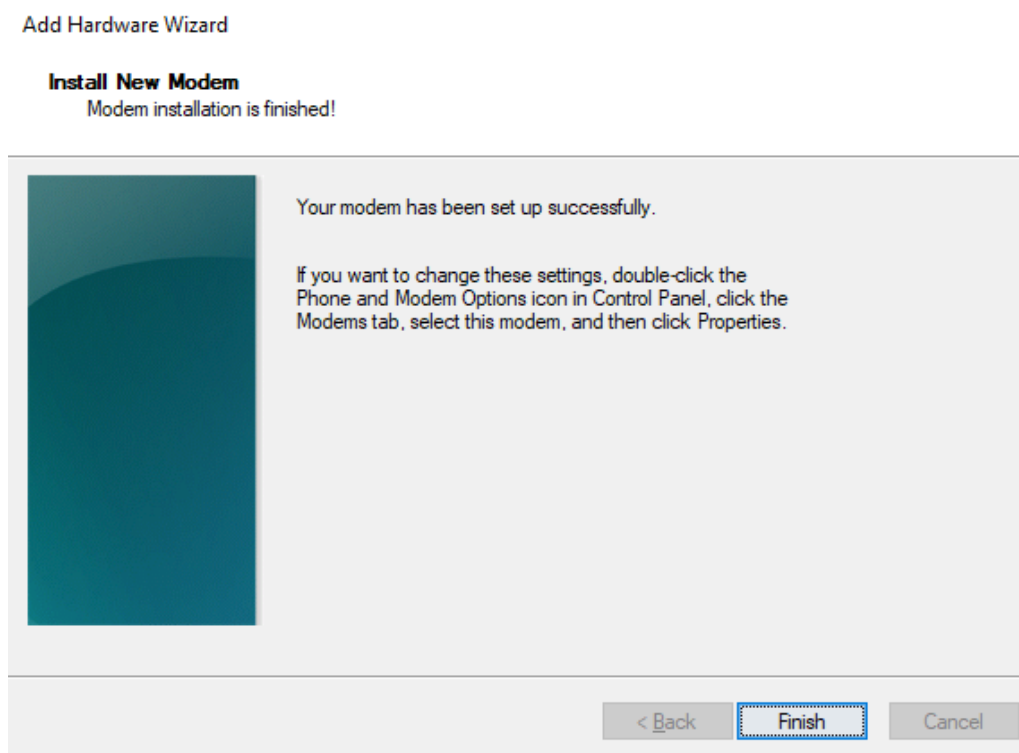


Figure 8: New Modem is Installed Successfully

6.2.1.2. Configure Modem Driver

Select the installed “**Standard 19200 bps Modem**” and click the “**Properties**” button. After entering the interface, select the “**Modems**” option and modify the “**Maximum Port Speed**” to “115200” (default value). Click the “**Advanced**” option, configure “**Extra Settings**” and input **AT+CGDCONT=1,"IP","CMNET"**.

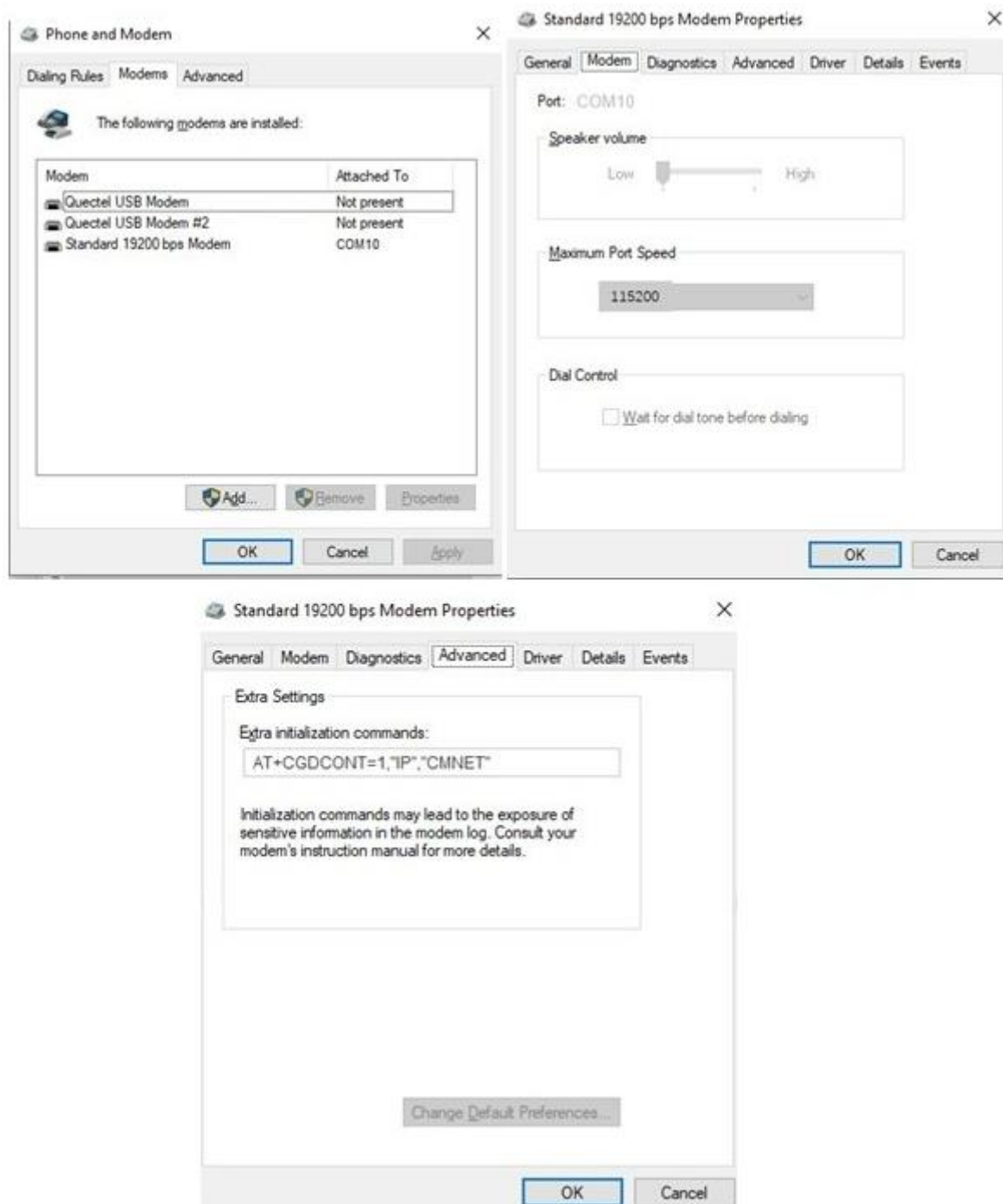


Figure 9: Configure Modem Driver

NOTE

In the example above, **AT+CGDCONT=1,"IP","CMNET"** predefines a PDP context where CID=1, PDP type=IP and APN=CMNET. CMNET is the APN of the network operator China Mobile and it should be replaced with the value provided by the network operator.

6.2.2. Dial-up Network Configuration

6.2.2.1. Create a New Connection

1. Open “Control Panel”, click “Network and Internet”, “Network and Sharing Center” and finally “Set up a new connection or network”.

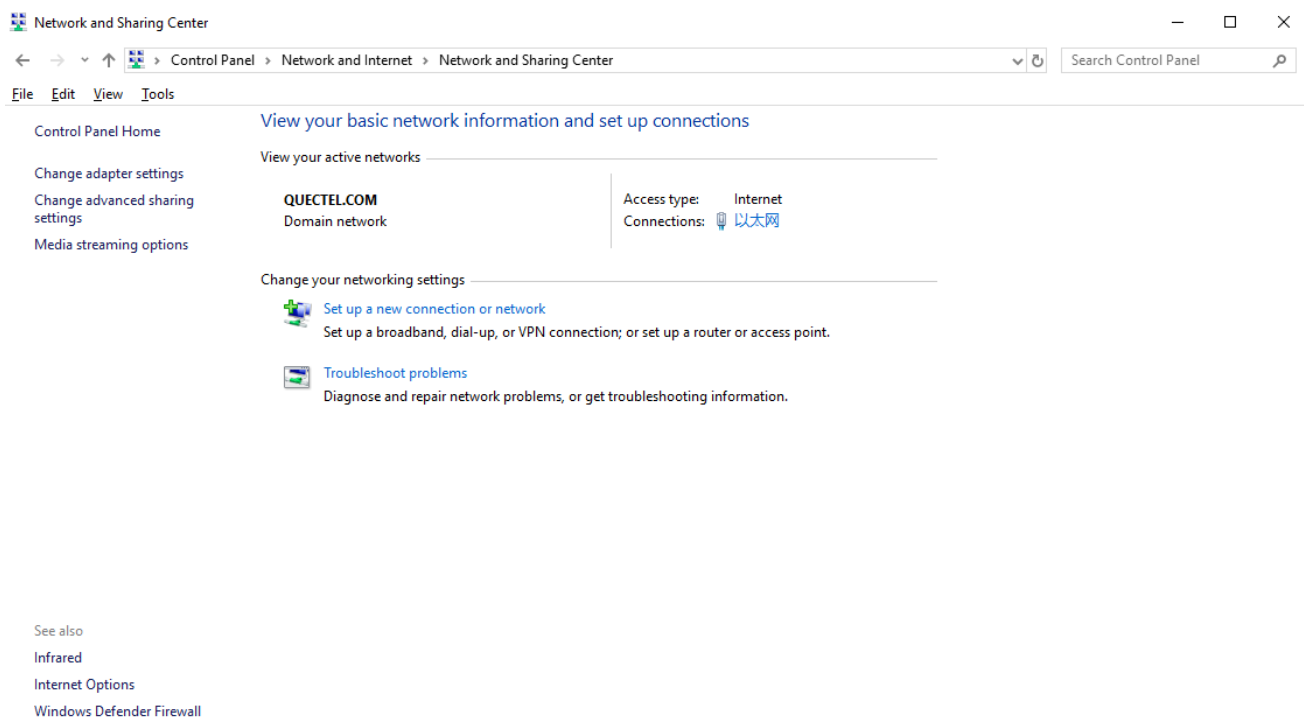


Figure 10: Create New Connection

2. Select **“Connect to the Internet”**, click **“Next”**. Then click **“Set up a new connection anyway”** and **“Dial-up”** to connect.

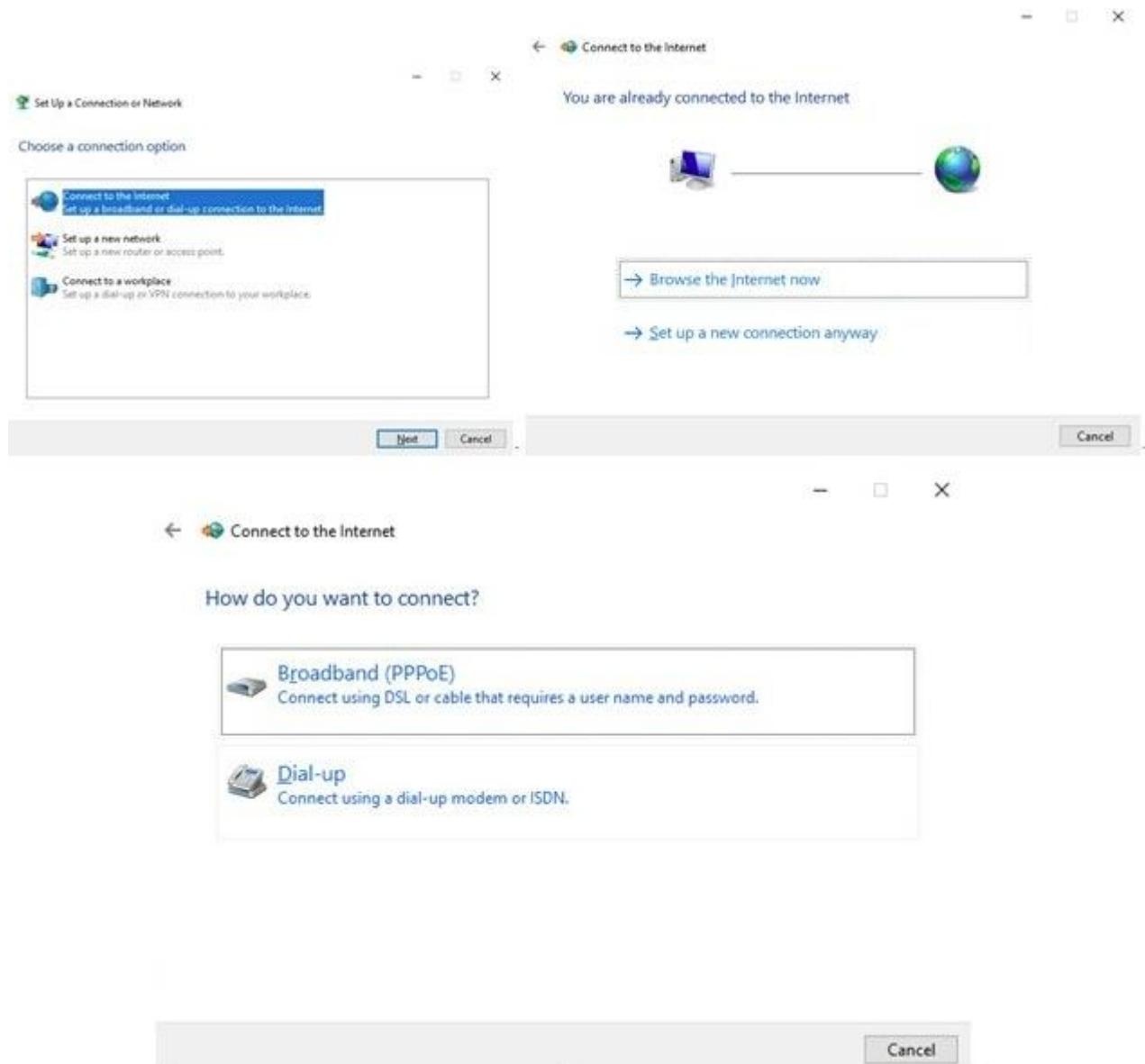
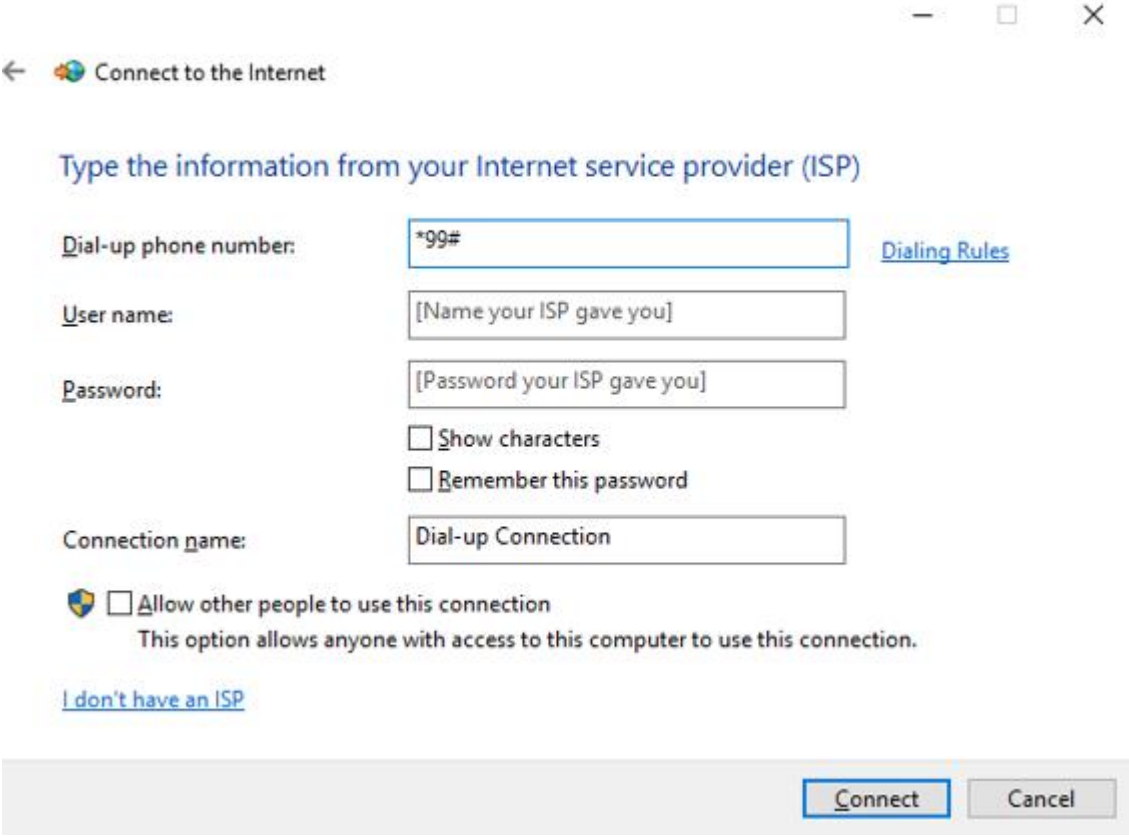


Figure 11: Set up New Connection

6.2.2.2. Configure the Connection

Enter characters (such as *99#) in “Dial-up phone number” box, and then click “**Connect**” button.



The screenshot shows a window titled "Connect to the Internet" with a back arrow icon on the left and standard window controls (minimize, maximize, close) on the right. The main heading is "Type the information from your Internet service provider (ISP)". Below this, there are several input fields and checkboxes:

- Dial-up phone number:** A text box containing "*99#" with a blue border. To its right is a blue link labeled "Dialing Rules".
- User name:** A text box containing "[Name your ISP gave you]".
- Password:** A text box containing "[Password your ISP gave you]". Below it are two checkboxes:
 - ☐ Show characters
 - ☐ Remember this password
- Connection name:** A text box containing "Dial-up Connection".
- Below the connection name is a checkbox with a shield icon: ☐ Allow other people to use this connection. Below this checkbox is the text: "This option allows anyone with access to this computer to use this connection."
- At the bottom left is a blue link: "I don't have an ISP".
- At the bottom right are two buttons: "Connect" (highlighted with a blue border) and "Cancel".

Figure 12: Configure Connection

6.2.2.3. Configure the Dial-up Tool

Enter characters (such as *99#) in “Dial” window, click “Properties” button, and configure the “Standard 19200 bps Modem (COM10)”. Then select “115200” from the drop-down list of “Maximum speed”. Click “OK” button to finish the configuration.

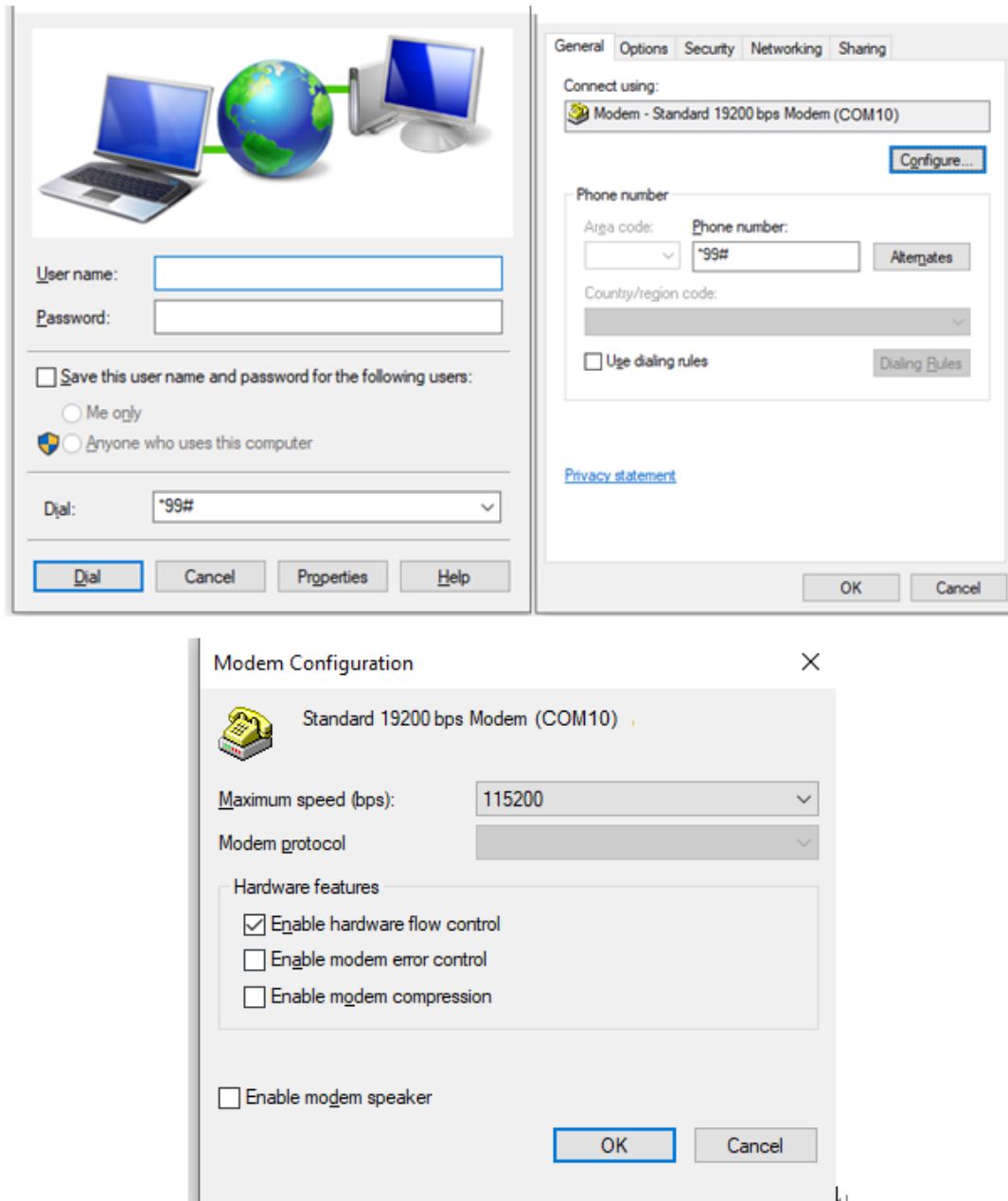


Figure 13: Configure Dial-up Tool

6.2.2.4. Establish the Dial-up Connection

After the above operations, click the “**Dial**” button, the interface will pop up a prompt box “**Verifying username and password**”, and the pop-up box quickly pops up “**Connected**” to indicate a dial-up connection. See the following figures for details.

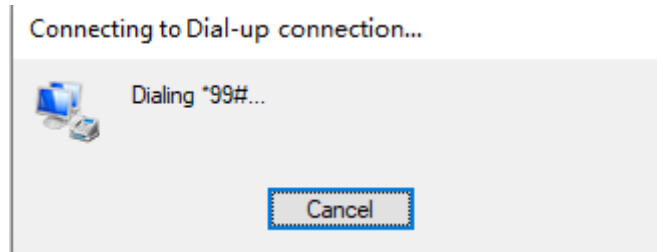


Figure 14: Connecting to Dial-up Connection

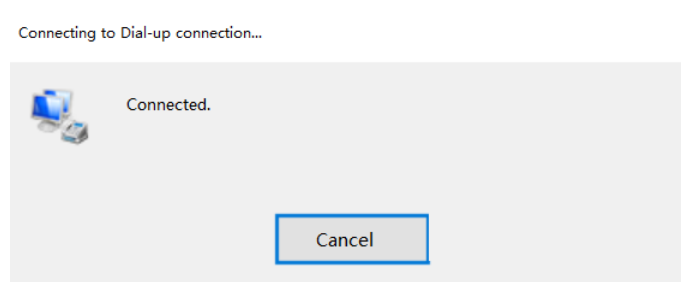


Figure 15: Establish Dial-up Connection Successfully

7 Appendix References

Table 2: Related Document

Document Name
[1] Quectel_UMTS_LTE_5G_Linux_USB_Driver_User_Guide

Table 3: Terms and Abbreviations

Abbreviation	Description
APN	Access Point Name
CHAP	Challenge Handshake Authentication Protocol
DCD	Data Carrier Detection
DNS	Domain Name Server
DTR	Data Terminal Ready
GPRS	General Packet Radio Service
GSM	Global System of Mobile Communication
IP	Internet Protocol
IPCP	IP Control Protocol
LCP	Link Control Protocol
MCU	Micro Control Unit
MS	Mobile Station
PAP	Password Authentication Protocol
PDP	Packet Data Protocol
PIN	Personal Identification Number

PPP	Point-to-Point Protocol
RI	Ring Indicator
TA	Terminal Adapter
TE	Terminal Equipment
UART	Universal Asynchronous Receiver Transmitter
(U)SIM	(Universal) Subscriber Identity Module
